DATABASE (SECP2523) – SEM I 2026/2027
ACTIVITY 1 – WEEK 1

1. Case Study Scenario

Universiti Teknologi Malaysia (UTM) plans to develop a Course Registration System to manage student information, course details and student course registrations. Students may register for multiple courses per semester, and a course may have multiple registered students. As database analysts, your group must identify suitable relational keys and explain how these keys support accurate data management.

The database contains three relations: STUDENT (student information), COURSE (course details) and REGISTRATION (student course registrations).

2. Relational Tables

Table 1. STUDENT

studentID	studentName	email	programme	yearOfStudy
A101	Ahmad Hakimi	ahmad@utm.my	SECPH	2
A102	Siti Aisyah	siti@utm.my	SECJH	1
A103	Kumar Raj	kumar@utm.my	SECPH	3
A104	Nur Amira	amira@utm.my	SECRH	2
A105	Daniel Lee	daniel@utm.my	SECJH	3
A106	Farah Nadia	farah@utm.my	SECPH	1

Table 2. COURSE

courseID	courseName	creditHour	department
SECP2523	Database	3	ACAI
SECP1513	Programming Technique I	3	ACAI
SECP2753	Data Mining	3	ACAI
SECI1013	Discrete Structure	3	Computing
SECR1013	Digital Logic	3	Computing

Table 3. REGISTRATION

registrationID	studentID	courseID	academicSession	semester	registrationStatus
R001	A101	SECP2523	2026/2027	1	Registered
R002	A101	SECP2753	2026/2027	1	Registered
R003	A102	SECP1513	2026/2027	1	Registered
R004	A103	SECP2523	2026/2027	1	Registered
R005	A103	SECP2753	2026/2027	1	Registered
R006	A104	SECI1013	2026/2027	1	Registered
R007	A105	SECR1013	2026/2027	1	Registered
R008	A106	SECP1513	2026/2027	1	Registered
R009	A102	SECP2523	2026/2027	1	Registered
R010	A105	SECP2523	2026/2027	1	Registered

Assumptions: studentID and courseID are permanent unique identifiers; registrationID is unique; email is unique and mandatory; a student can register for a given course only once within a particular academic session and semester.

Task 1: Identify Relational Keys

Working in your group, identify the candidate key(s), primary key, alternate key(s), foreign key(s), and composite key(s) for each relation. Complete the table and justify your choices during discussion.

Relational Key	STUDENT	COURSE	REGISTRATION
Candidate Key(s)	studentID, email	courseID	RegistrationID, courseID, studentID academicsession, semester
Primary Key	studentID	courseID	registrationID
Alternate Key(s)	email	none	studentID, courseID, academicsession semester
Foreign Key(s)	none	none	studentID, courseID
Composite Key(s)	none	none	studentID + courseID + academicsession + semester

Task 2: Analyse Relationships

1. Which attribute uniquely identifies each student in STUDENT?
2. Why is studentName not suitable as a primary key?
3.
4. Can email be used as a candidate key? Justify your answer.
5. Which attribute uniquely identifies each course in COURSE?
6. Identify the foreign keys in REGISTRATION and their referenced tables.
7. Why cannot studentID alone be used as the primary key in REGISTRATION?
8. Identify a possible composite candidate key for REGISTRATION.
9. Can a student register for the same course in different semesters? Explain how the database supports this requirement.

1) studentID

2) Because others student might have the same name. Plus the context of the table is student. Students name are also not guaranteed to be unique

4) Yes, because one email can not be the same to others. Plus, email is also unique to every student

5) courseID

6) Foreign keys : studentID from table student
courseID from table course

7) A single student can register multiple course Plus, repeating studentID values for multiple course registration would violate primary key uniqueness

8) studentID + courseID + academicsession + semester

9) Yes, because when the semester or academicsession change the composite keys remains unique.

Task 3: Database Key Validation

Determine whether each proposed operation is valid or invalid, and justify your decision. Assume that unspecified fields are valid and the registration period is open.

No.	Proposed Operation	Valid / Invalid	Justification
1	Insert a new student with studentID A107.	valid	does not exist yet
2	Insert another student using studentID A101.	Invalid	already exist
3	Register A101 for SECP1513 in Semester 1, 2026/2027, using R011.	valid	That ID has not registered for that course yet
4	Register A999 for SECP2523 using R012.	Invalid	A999 does not exist in the student table
5	Insert a new course using courseID SECP2523.	Invalid	already exist
6	Insert another registration using registrationID R001.	Invalid	already exist
7	Register A101 again for SECP2523 in Semester 1, 2026/2027, using R013.	Invalid	can not registered same course twice in the same semester and session
8	Register A101 for SECP2523 in Semester 2, 2026/2027, using R014.	valid	the semester value is different